RESULTS REPORT

Model training, evaluation and findings

What was trained, how it was measured, and what it showed

DM-AHR-ViT · Explainable Vision Transformer for Skin Lesion Classification Using Dermoscopy Images

Research prototype only. Not clinically validated and not intended for diagnosis, triage or patient care.

In one sentence

Across 15 trained models (5 arms × 3 seeds) evaluated once on a locked test set of 1,502 images, the best arm was A — original at macro-F1 0.750. Our own diffusion arm C — DM-AHR diffusion (0.426) was far behind, and the authors' released checkpoint (Creleased, 0.713) closed most of that gap but still did not beat classical DullRazor or original images. The honest conclusion is 'hair removal did not help this classifier', not 'DM-AHR cannot remove hair' — the released checkpoint does remove hair, it just does not improve classification.

1. What was trained and how

Fifteen Vision Transformers were trained: five input conditions (arms) at three random seeds each. Everything except the arm is identical — same frozen split, same manifests, same architecture, hyperparameters, augmentation and evaluation code — so a difference between arms is attributable to the hair-removal condition alone. Creleased is the authors' released DM-AHR checkpoint run through our pipeline at a 25-step respaced schedule; it was added after the protocol was frozen and is reported as exploratory.

Arm	Input the classifier sees	Macro-F1 (mean ± SD)
A — original	untouched dermoscopy image	0.750 ± 0.005
B — DullRazor classical	classical hair removal (DullRazor-style)	0.698 ± 0.009
C — DM-AHR diffusion	diffusion-generated hairless image	0.426 ± 0.013
A* — 128 px control	original through the same 128 px round-trip	0.732 ± 0.008
Creleased — released ckpt	released DM-AHR checkpoint, 25-step respaced (exploratory)	0.713 ± 0.028

Training recipe (identical for all fifteen runs)

Component	Setting
Backbone	ViT-B/16, ImageNet-1k pretrained, 224 px input, 196 patches
Optimiser	AdamW, lr 3e-5, weight decay 0.05, warm-up 5 epochs then cosine decay
Batch / precision	32, bf16 autocast, gradient clipping 1.0
Class imbalance	inverse-frequency class weights from the training split only
Early stopping	validation macro-F1, patience 10, max 100 epochs
Seeds	42, 123, 2026
Test protocol	the locked test set is scored once per frozen checkpoint

Compute environment

All GPU work ran on Modal: NVIDIA A100 80GB PCIe, Python 3.11.12, torch 2.5.1+cu124, CUDA 12.4. A typical training run took about 6.5 minutes (25 epochs to early stopping). Per-run GPU type and wall-clock are logged in artifacts/logs/run_registry.jsonl.

2. Classification results

All metrics are computed on the held-out test set of 1,502 images. Macro-F1 is the primary endpoint because classes are severely imbalanced: nv is two thirds of the data, so plain accuracy rewards ignoring the rare classes.

Arm	Macro-F1	Accuracy	Balanced acc.	Macro AUC
A — original	0.750 ± 0.005	0.858 ± 0.012	0.735	0.969

Arm	Macro-F1	Accuracy	Balanced acc.	Macro AUC
A* — 128 px control	0.732 ± 0.008	0.851 ± 0.006	0.726	0.969
B — DullRazor classical	0.698 ± 0.009	0.842 ± 0.005	0.687	0.965
C — DM-AHR diffusion	0.426 ± 0.013	0.685 ± 0.019	0.452	0.878
Creleased — released ckpt	0.713 ± 0.028	0.848 ± 0.009	0.696	0.967

Hair-present subset

Arm	Macro-F1	n
A — original	0.749 ± 0.002	1316
A* — 128 px control	0.735 ± 0.013	1316
B — DullRazor classical	0.691 ± 0.009	1316
C — DM-AHR diffusion	0.411 ± 0.017	1316
Creleased — released ckpt	0.712 ± 0.029	1316

This is the subset where hair removal actually operates — the images it could help. The ordering is unchanged from the full test set.

Per-class test metrics

Class	Arm	Support	F1	Recall (sensitivity)	Specificity	AUC
akiec	A	49	0.625	0.694	0.982	0.956
	B	49	0.589	0.639	0.982	0.947
	C	49	0.286	0.408	0.952	0.911
	Astar	49	0.556	0.673	0.974	0.961
	Creleased	49	0.562	0.633	0.979	0.954
bcc	A	77	0.750	0.706	0.991	0.975
	B	77	0.724	0.740	0.984	0.979
	C	77	0.315	0.290	0.971	0.881
	Astar	77	0.747	0.719	0.989	0.982
	Creleased	77	0.726	0.736	0.984	0.976
bkl	A	165	0.676	0.644	0.967	0.955
	B	165	0.628	0.590	0.965	0.947
	C	165	0.356	0.461	0.863	0.791
	Astar	165	0.677	0.630	0.971	0.949
	Creleased	165	0.692	0.679	0.965	0.946
df	A	17	0.733	0.706	0.998	0.987
	B	17	0.591	0.529	0.997	0.978
	C	17	0.140	0.137	0.990	0.863
	Astar	17	0.719	0.627	0.999	0.985
	Creleased	17	0.620	0.471	1.000	0.980

Class	Arm	Support	F1	Recall (sensitivity)	Specificity	AUC
mel	A	167	0.634	0.615	0.959	0.939
	B	167	0.561	0.529	0.957	0.935
	C	167	0.400	0.373	0.939	0.820
	Astar	167	0.613	0.611	0.952	0.925
	Creleased	167	0.609	0.617	0.949	0.936
nv	A	1006	0.943	0.956	0.855	0.981
	B	1006	0.942	0.958	0.844	0.980
	C	1006	0.871	0.827	0.855	0.922
	Astar	1006	0.943	0.949	0.872	0.982
	Creleased	1006	0.942	0.941	0.884	0.980
vasc	A	21	0.889	0.825	1.000	0.991
	B	21	0.853	0.825	0.998	0.991
	C	21	0.615	0.667	0.993	0.955
	Astar	21	0.868	0.873	0.998	0.999
	Creleased	21	0.840	0.794	0.999	0.994

Recall is the proposal's sensitivity. Supports for df (17) and vasc (21) are small, so their per-class numbers are noisy and are not interpreted individually.

3. Hair-removal image quality

Measured on the 186 held-out synthetic pairs — hair composited onto genuinely hair-absent images, so a true hairless reference exists. The unprocessed hairy input is scored as a floor: falling below it means the method made the image worse than doing nothing.

Condition	PSNR (dB) mean ± SD	SSIM mean ± SD
Hairy input (floor)	22.13 ± 1.90	0.520 ± 0.093
Arm B — classical	35.73 ± 2.11	0.918 ± 0.023
Creleased — released ckpt	27.08 ± 2.63	0.703 ± 0.094
Arm C — diffusion	14.29 ± 1.09	0.074 ± 0.006

Why Arm C scores below the untouched input — diagnosed

A diagnosis on validation pairs separates the two candidate causes. A hairless reference round-trips through the same inference path at PSNR 34.6 / SSIM 0.990, and the released DM-AHR checkpoint reaches PSNR 26.8 / SSIM 0.684 through that path — above the hairy-input floor of 22.1. Our checkpoint reaches PSNR 15.0 / SSIM 0.079, below the floor. The adapter is sound; the checkpoint is undertrained. Its output is not over-smoothed: it carries roughly 15× the reference's high-frequency energy with only a modest drop in global contrast, which is what a model that has not learned the mapping looks like. It is labelled an under-resourced ablation throughout and is not presented as DM-AHR at published settings. See reports/arm_c_diagnosis.md.

4. Explainability

For every test image, two saliency methods — Grad-CAM on the patch grid and attention roll-out — were compared against the released HAM10000 lesion mask. The table reports the fraction of the top-20% most salient pixels that fall inside the mask, averaged over seeds, with the two chance baselines for context.

Arm	Grad-CAM overlap	Roll-out overlap	Grad-CAM pointing	Roll-out pointing	Uniform chance	Gaussian chance
A — original	0.363	0.555	0.468	0.842	0.200	0.644
A* — 128 px control	0.404	0.577	0.536	0.858	0.200	0.644
B — DullRazor classical	0.388	0.586	0.494	0.876	0.200	0.644
C — DM-AHR diffusion	0.412	0.337	0.617	0.242	0.200	0.644
Creleased — released ckpt	0.382	0.568	0.524	0.878	0.200	0.644

Overlap measures spatial alignment with released masks, not clinically correct reasoning. Every arm beats uniform chance, but a centred-Gaussian prior scores 0.644 on overlap and 0.953 on the pointing game, exceeding every arm on both measures. No arm therefore beats a trivial attend-to-the-centre baseline, and these numbers do not establish that any arm localises lesions better than that prior. The masks themselves are single-rater: a dermatologist (Tschandl P) created binary masks for all 10,015 images, network-initialised and then verified, corrected or replaced by hand in FIJI (Tschandl et al., Nature Medicine 2020), so overlap is measured against one tracing, not consensus.

[figure unavailable: dmahr.jpg]

5. Statistical comparisons

Every endpoint is tested twice, as the protocol requires: a lesion-group bootstrap (10,000 replicates, the same resample shared across arms and seeds, percentile 95% CI) and a Wilcoxon signed-rank test on values collapsed to one per lesion group. For the classification endpoint the per-group value is that group's mean per-image correctness, so the Wilcoxon test asks whether the arms differ in per-lesion accuracy — it is not a test of macro-F1 itself, and the column header says so. Holm correction is applied across the three planned comparisons within each endpoint, separately for each test; A–A* is a secondary control in its own family and cannot inflate the planned family. A CI spanning zero is called inconclusive, not 'no difference'.

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Wilcoxon p (per-group acc.)	Groups
paired_psnr	B-hairy input	+13.604	+13.261 to +13.943	<0.0001	<0.0001	186
paired_psnr	C-B	-21.441	-21.768 to -21.106	<0.0001	<0.0001	186
paired_psnr	C-hairy input	-7.838	-8.088 to -7.592	<0.0001	<0.0001	186

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Wilcoxon p (per-group acc.)	Groups
paired_ssim	B-hairy input	+0.398	+0.385 to +0.410	<0.0001	<0.0001	186
paired_ssim	C-B	-0.843	-0.847 to -0.840	<0.0001	<0.0001	186
paired_ssim	C-hairy input	-0.446	-0.459 to -0.433	<0.0001	<0.0001	186
test_macro_f1	A-Astar	+0.018	-0.008 to +0.045	0.1668	0.0739	1437
test_macro_f1	A-B	+0.052	+0.019 to +0.087	0.0012	0.0113	1437
test_macro_f1	A-C	+0.324	+0.283 to +0.366	<0.0001	<0.0001	1437
test_macro_f1	B-C	+0.272	+0.225 to +0.320	<0.0001	<0.0001	1437
test_macro_f1	C-Astar	-0.306	-0.348 to -0.265	<0.0001	<0.0001	1437
test_macro_f1	A-Astar	+0.014	-0.014 to +0.041	0.3218	0.0800	1251
test_macro_f1	A-B	+0.058	+0.022 to +0.095	0.0008	0.0058	1251
test_macro_f1	A-C	+0.337	+0.291 to +0.383	<0.0001	<0.0001	1251
test_macro_f1	B-C	+0.279	+0.224 to +0.333	<0.0001	<0.0001	1251
test_macro_f1	C-Astar	-0.324	-0.370 to -0.279	<0.0001	<0.0001	1251
x1_gradcam_overlap20	A-B	-0.027	-0.039 to -0.017	<0.0001	<0.0001	1437
x1_gradcam_overlap20	A-C	-0.053	-0.065 to -0.040	<0.0001	<0.0001	1437
x1_gradcam_overlap20	B-C	-0.025	-0.037 to -0.013	0.0002	<0.0001	1437
x1_rollout_overlap20	A-B	-0.032	-0.034 to -0.029	<0.0001	<0.0001	1437
x1_rollout_overlap20	A-C	+0.217	+0.208 to +0.226	<0.0001	<0.0001	1437
x1_rollout_overlap20	B-C	+0.249	+0.239 to +0.258	<0.0001	<0.0001	1437

Exploratory: released-checkpoint arm

These rows are not part of the pre-registered family. Creleased uses the authors' released DM-AHR checkpoint at a 25-step respaced schedule, a deviation from the paper's 2,000 steps, and its classifier was trained on our budget. Holm correction is applied within this exploratory family only.

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Cohen's dz	Groups
paired_psnr	Creleased-B	-8.657	-9.063 to -8.254	<0.0001	-3.073	186
paired_psnr	Creleased-hairy input	+4.946	+4.503 to +5.388	<0.0001	+1.639	186
paired_ssim	Creleased-B	-0.215	-0.228 to -0.204	<0.0001	-2.558	186

Endpoint	Comparison	Diff.	95% CI (bootstrap)	p (Holm)	Cohen's dz	Groups
paired_ssim	Creleased-hairy input	+0.182	+0.166 to +0.199	<0.0001	+1.595	186
test_macro_f1	A-Creleased	+0.038	+0.005 to +0.071	0.0476	+0.053	1437
test_macro_f1	B-Creleased	-0.015	-0.049 to +0.019	0.4004	-0.011	1437
test_macro_f1	A-Creleased	+0.037	+0.002 to +0.073	0.0768	+0.060	1251
test_macro_f1	B-Creleased	-0.021	-0.057 to +0.016	0.2614	-0.016	1251

6. What this says about the three hypotheses

H1 — image quality (diffusion preserves texture better than classical):

- PSNR: B higher than C (C - B = -21.441, 95% CI -21.768 to -21.106). SSIM: B higher than C (C - B = -0.843, 95% CI -0.847 to -0.840). H1 is not supported by these data. Arm C's SSIM is near-constant across all 186 pairs (0.074 ± 0.006, with 186 below 0.15), which fits an undertrained model. The released checkpoint was then scored through the same adapter (Creleased): it clears the hairy-input floor by 4.95 dB but sits 8.66 dB below classical inpainting, so H1 is now tested for DM-AHR and is not supported.

H2 — classification (diffusion-cleaned images improve the ViT):

- A vs C: A higher than C (A - C = +0.324, 95% CI +0.283 to +0.366). B vs C: B higher than C (B - C = +0.272, 95% CI +0.225 to +0.320). The direction is opposite to the hypothesis and the gap is large. The released checkpoint does not rescue it: A vs Creleased is +0.038 (95% CI +0.005 to +0.071, Holm p=0.048, negligible effect) and B vs Creleased is inconclusive (-0.015, CI -0.049 to +0.019, Holm p=0.400).

H3 — explainability (attention focuses more on lesions after diffusion):

- Grad-CAM overlap, A vs C: C higher than A (A - C = -0.053, 95% CI -0.065 to -0.040). Attention roll-out, A vs C: A higher than C (A - C = +0.217, 95% CI +0.208 to +0.226). The methods disagree in direction, but H3 is not supported and the measurement is uninformative as specified: a centred-Gaussian prior beats every arm on both overlap (0.644) and pointing (0.953), so no arm demonstrates lesion localisation beyond a centre bias. That is a limitation of the measurement, not a finding about diffusion hair removal, and it must be fixed before H3 can be tested.

Resolution control:

- A vs A*: inconclusive (A - A* 95% CI -0.008 to +0.045). Because this is inconclusive, the 128 px round-trip alone does not explain Arm C's deficit.

7. Limitations

- **Arm C is under-resourced, and the cause is diagnosed —** It ran 20,000 training iterations (the upstream config's n_iter is 660,000 and the released checkpoint is named I725000_E1083), 1,000 sampling steps against 2,000, and 128 px upsampled to 224. A hairless reference round-trips through the same inference path at PSNR 34.6 / SSIM 0.990, so the adapter is sound and the checkpoint is undertrained. Its results describe that model, not the published method.
- **Creleased is exploratory, not pre-registered —** It uses the authors' released checkpoint at a 25-step respaced schedule, a deviation from the paper's 2,000 steps,

and its classifier was trained on our budget rather than the paper's. Its comparisons are reported in their own Holm family and must not be read as pre-registered results.

- **Three seeds —** inference is over ~1,437 lesion groups with seeds averaged inside each replicate, never counted as independent samples; still, three seeds cannot characterise variance exhaustively.
- **Screening unvalidated —** the proposal specifies the manually annotated hair masks of Lama et al. (ChimeraNet) as an independent reference for screening quality, and those masks were not obtained. This is a skipped step, not a gap in the data. The threshold was calibrated on synthetic pairs and frozen, which protects the test set but does not validate the detector on real hair. The 89.8% screening prevalence is reported as a limitation.
- **Masks are single-rater —** the HAM10000 segmentations were created by one dermatologist (Tschandl P), network-initialised and then verified, corrected or replaced by hand in FIJI (Tschandl et al., Nature Medicine 2020). No inter-rater agreement is available, so explanation overlap is measured against one tracing.
- **Single corpus —** one public dataset; nothing here establishes behaviour on other scanners or populations.
- **Licensing unresolved —** HAM10000 is CC BY-NC 4.0; the DM-AHR repository's README (CC BY-NC 4.0) and LICENSE (Apache-2.0) contradict each other. Commercial use needs a rights decision.

8. Reproducibility

- **Determinism —** Arm A seed 42 re-run from scratch reproduced validation macro-F1 exactly (0.778336 vs 0.778336).
- **Portability —** the frozen split hash was reproduced byte for byte across two Modal accounts, and the full data-preparation chain was re-run natively with no Modal present, giving the identical hash.
- **How to explore —** notebooks/01_results_walkthrough.ipynb reads the committed artifacts and explains these tables interactively; notebooks/02 and 03 reproduce the pipeline on Modal or a local GPU.

Every table here is generated from artifacts/logs/: classification_seed_summary.csv, classification_per_class.csv, paired_psnr_ssim_test.csv, xai_localization_*.csv and statistical_comparisons.csv. This document is built by scripts/14_results_report.py and can be regenerated after any rerun.